vibrantlearn.

Features and demo guide

Web: AI-Powered Personalized Learning Path Generator (Smart EdTech Learning System). Student-side prototype v0.4.



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What VibrantLearn is

A student types what they need to study and by when, for example "Java internal tomorrow 10am, BCA SEP", or uploads their own notes. VibrantLearn turns that into a personal study path: a timed plan with checkpoints, short notes in English, Kannada or Hindi, likely exam questions, quizzes and flashcards.

Who it is for

Student	Situation	What VibrantLearn gives them
The crammer	Internal test in a few hours	A tight plan of only the highest-mark topics, in focus blocks
The planner	Semester exam in weeks	Week-by-week path with checkpoints and revision loops
The notes-holder	Has the lecturer's PDF or photos	Answers from their own notes, with the page shown
The regional-language learner	Understands better in Kannada or Hindi	Notes in their language, technical terms kept in English

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1. Running the demo

Everything is in one folder: `E:\Startup\VibrantLearn\Code`. Open it in any IDE (VS Code, IntelliJ, Cursor) and use either option. Node.js 20 or newer is the only requirement.

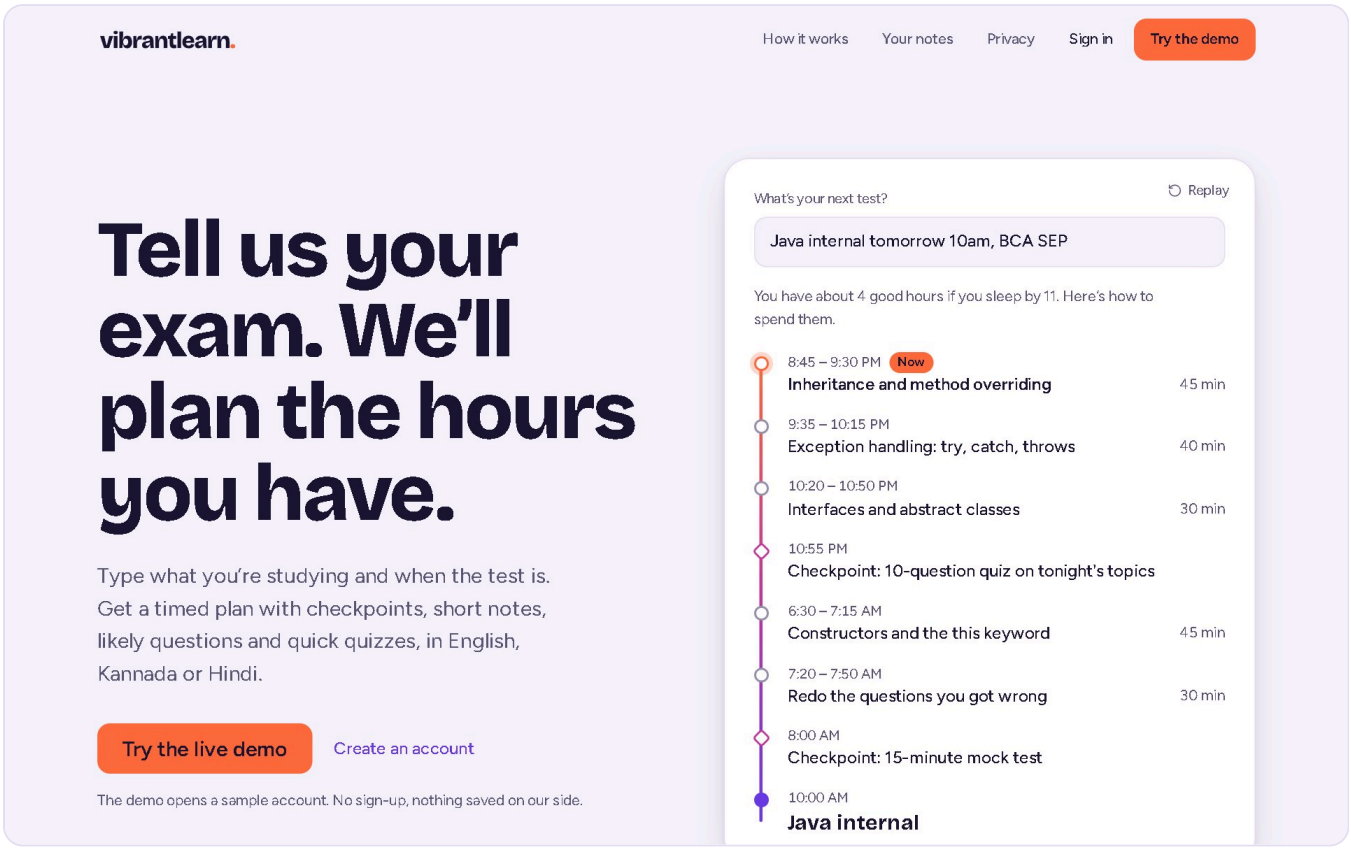
Option	How
Double-click	`START-DEMO.bat` in the folder
Any terminal	`npm run demo` from the folder

The first run installs packages and builds the app (a few minutes). After that it starts in seconds and opens `http://localhost:3000`. Press **Try the demo** to enter the app with a sample 3rd-semester BCA student. No account is needed for the demo; it keeps its data in the browser. Real accounts (email sign-in) save to the cloud database.

> To rebuild after code changes: `npm run demo:rebuild`. To run the full automatic check: `uv run --no-project --with playwright python apps/web/e2e/smoke.py` (17 steps).

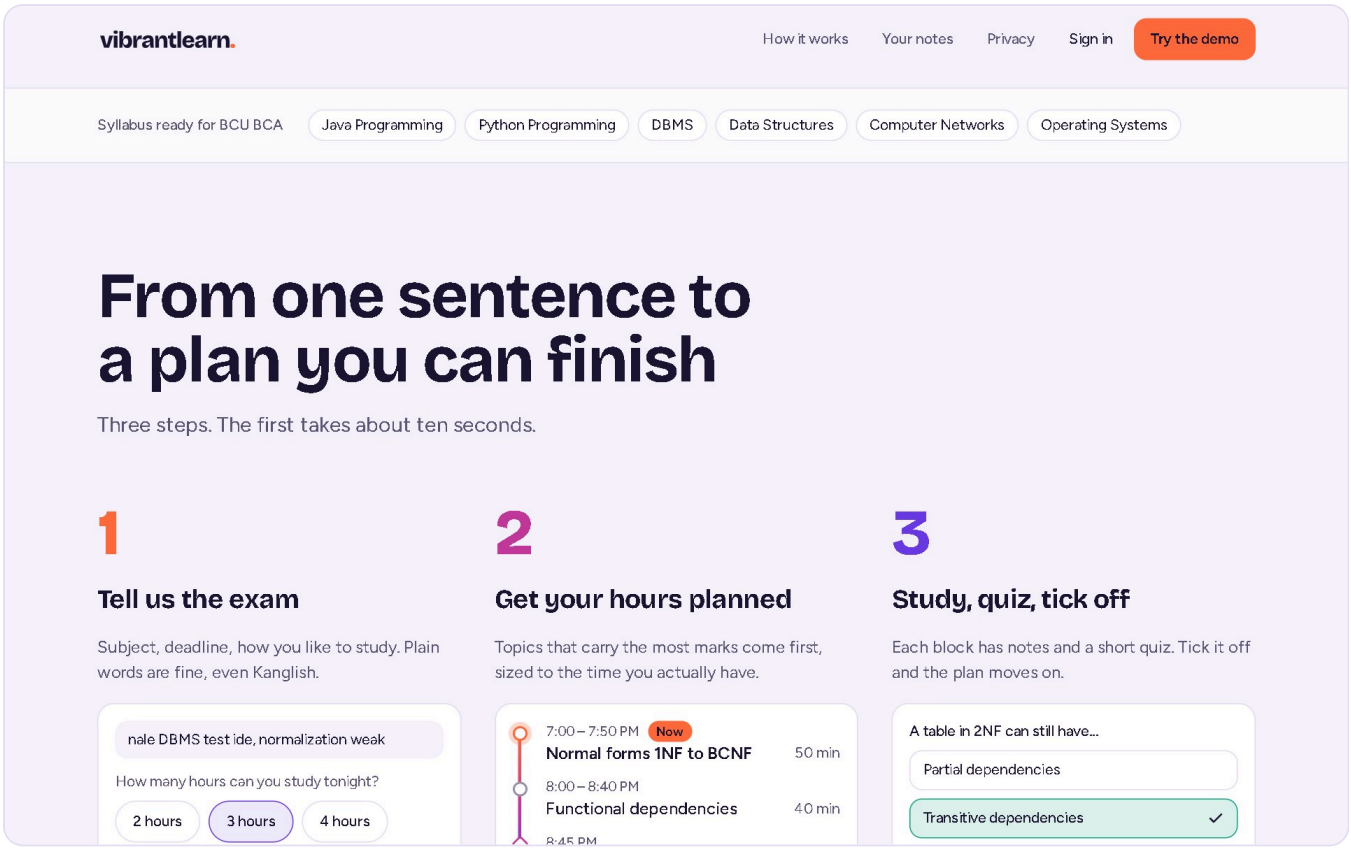
2. Landing page

The public page explains the product by showing it: a request is typed and a real study path draws itself next to the headline.

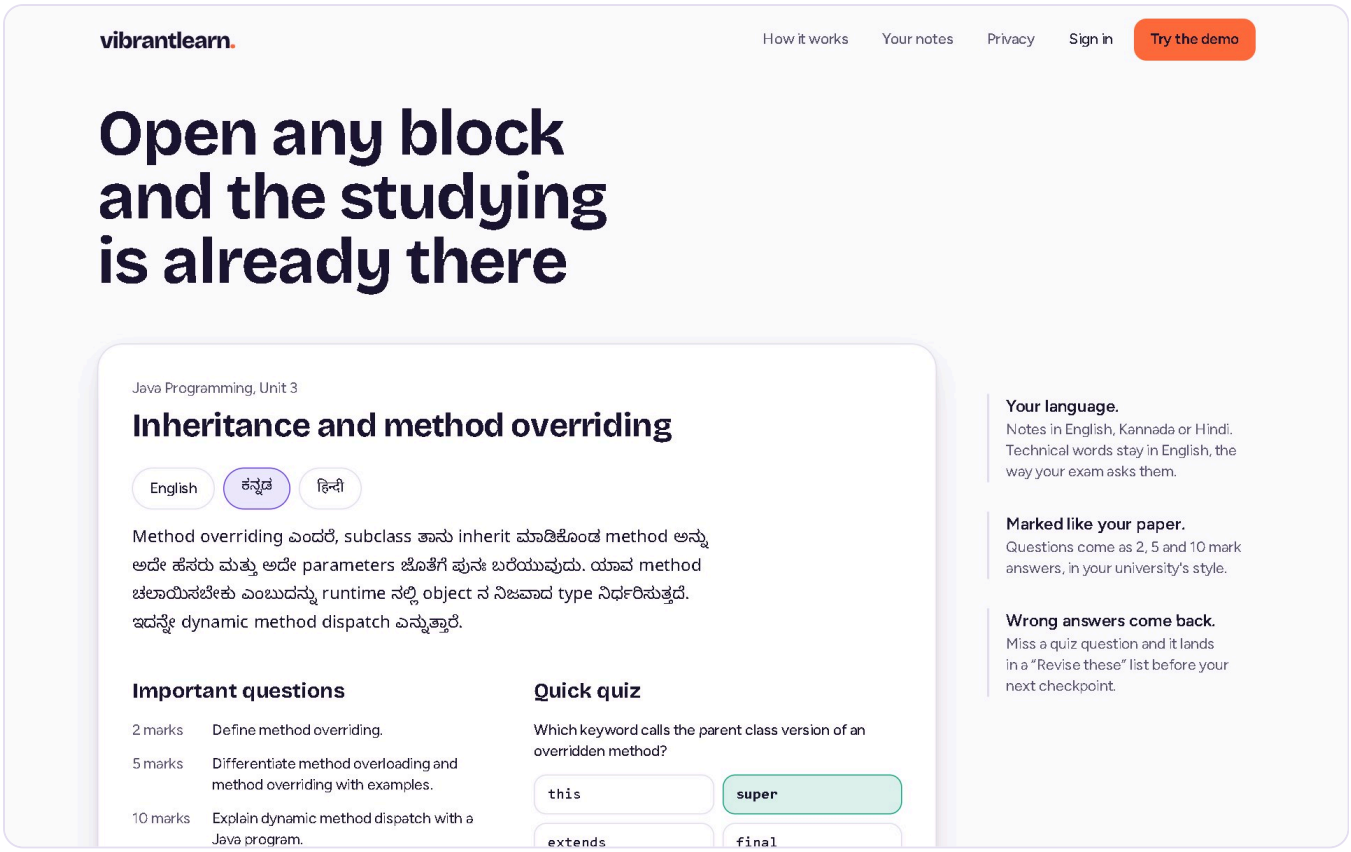


The hero: one sentence becomes a timed plan for a Java internal at 10 AM tomorrow.

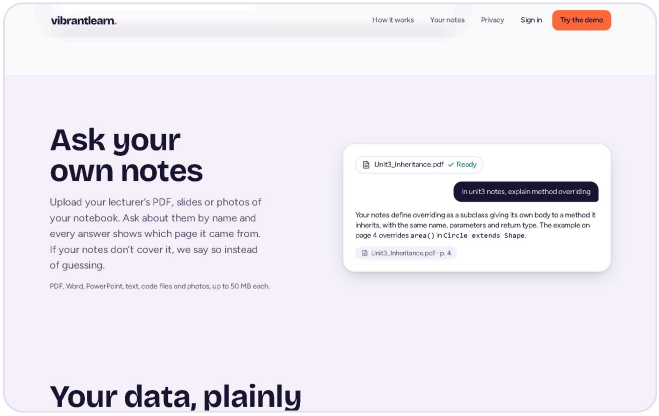
- **Live plan demo** in the hero, with a Replay button. People who prefer reduced motion see the finished plan straight away.
- **Scroll motion:** a brand-coloured progress line, sections that ease in, and a header that turns solid as you scroll.
- **One-click demo** from the header or hero. "Create an account" creates a real account in the cloud database.



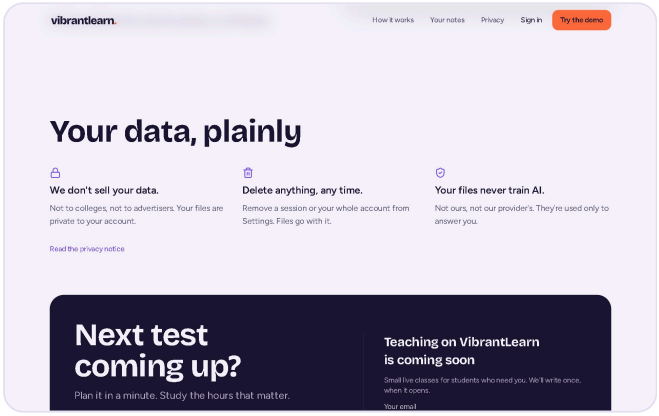
How it works: three real steps, each shown with the actual app fragment.



"Open any block": notes switchable between English, ಕನ್ನಡ and हिन्दी, 2/5/10-mark questions and a quick quiz.



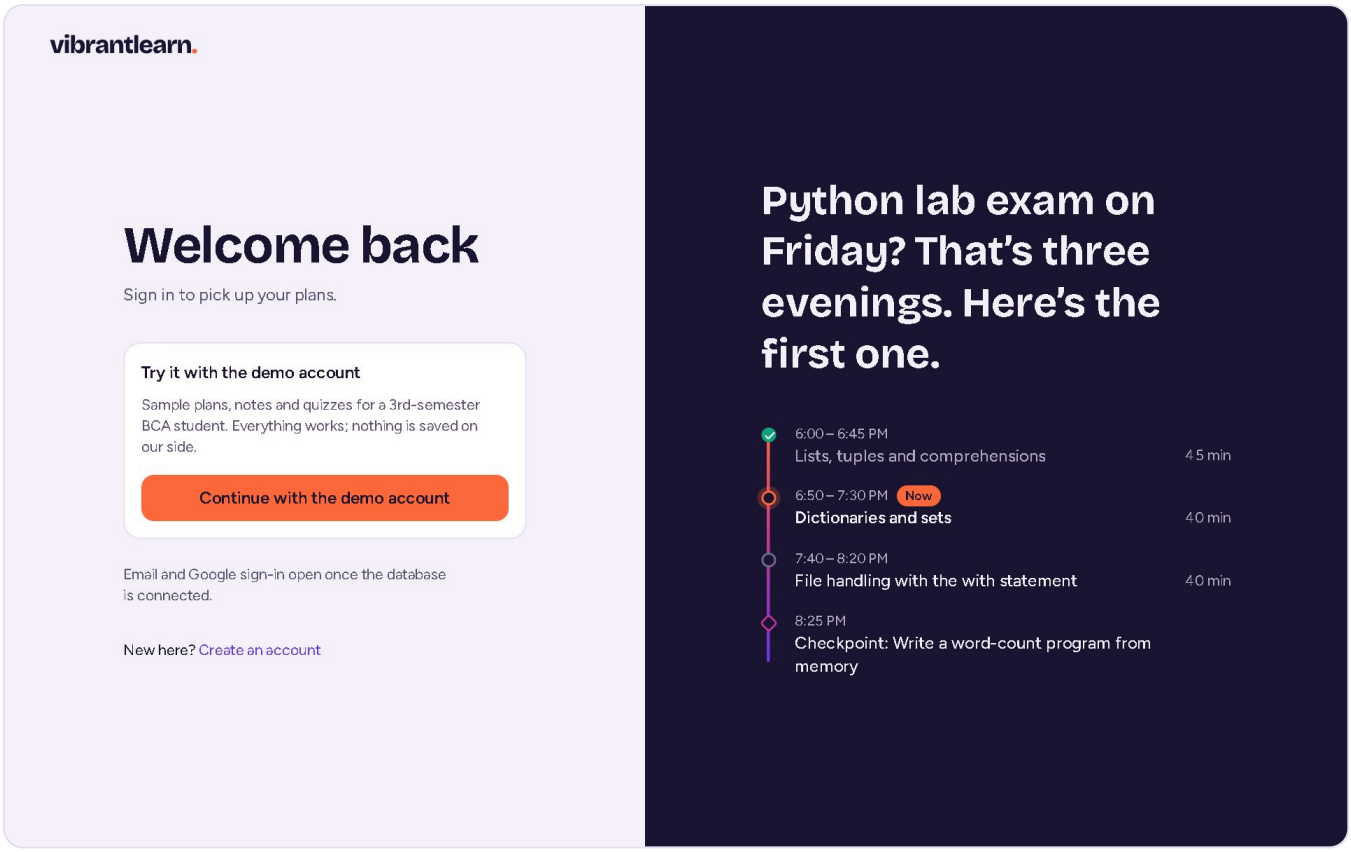
Ask your own notes, with the source page shown.



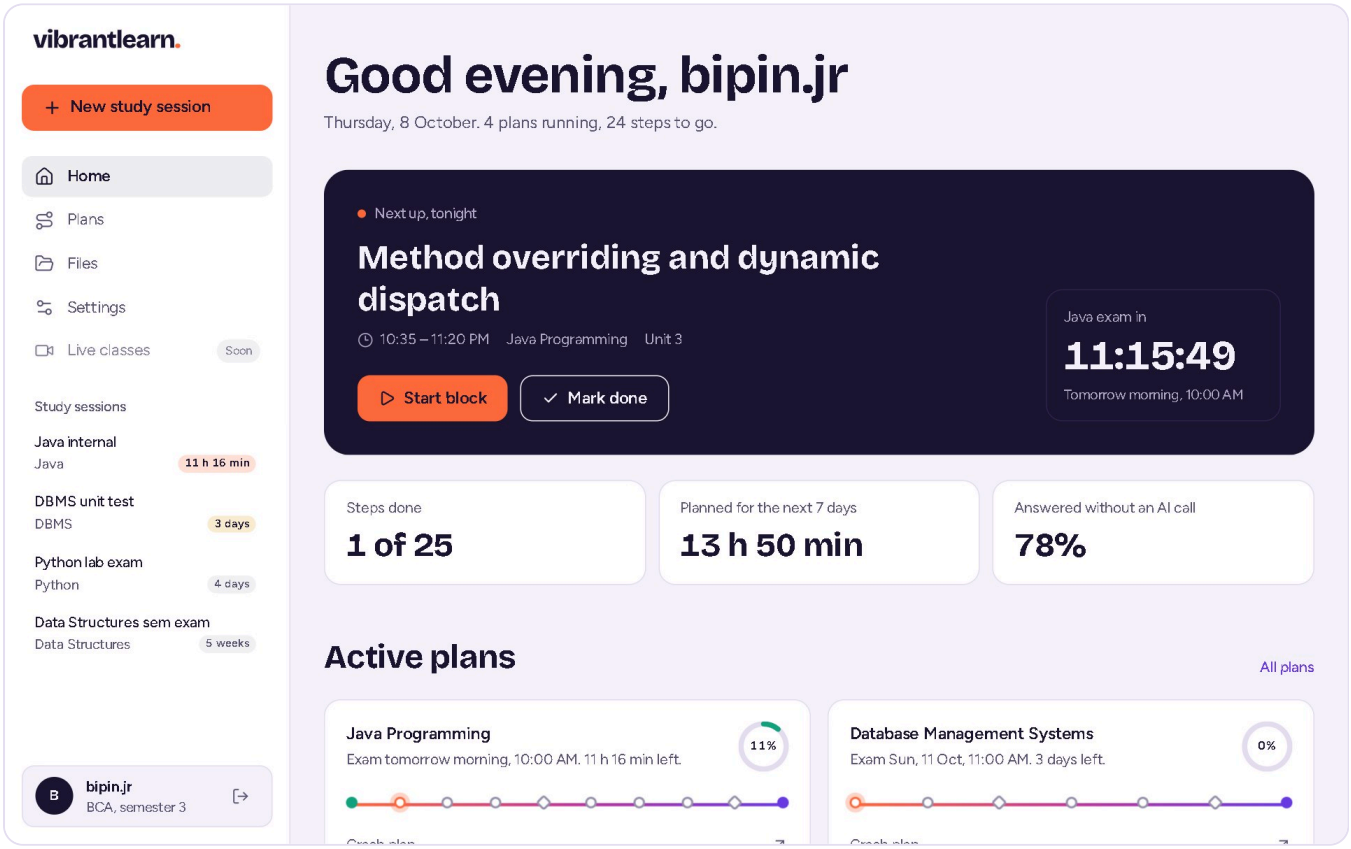
Privacy in plain words, and the teacher waitlist.

- No invented numbers or fake testimonials. Every example uses real BCA syllabus content.
- The footer credits the team: built by Bipin A and Darshan H.S.

3. Demo sign-in and dashboard



Sign-in page. In demo mode one button opens the app; email sign-in creates a real account saved in the cloud database.

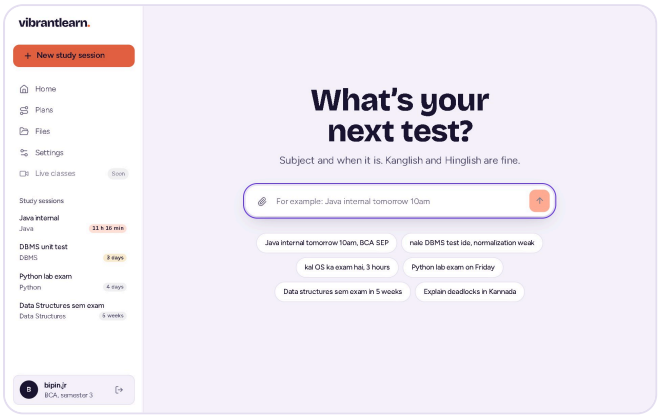


Dashboard: the next study block, a live countdown to the exam, progress and all active plans.

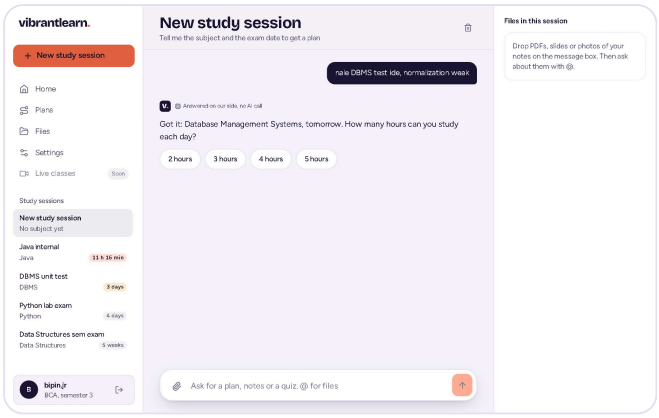
- **Next up** shows exactly what to study now, with Start block and Mark done.
- **Live countdown** to the exam, ticking every second.
- **Real progress numbers** worked out from the plans: steps done, hours planned this week, and how many answers needed no AI call.
- **Sidebar** lists every study session with time left, coloured by urgency (under a day, under three days, later).

4. Asking in plain language

Students write the way they talk, including Kanglish and Hinglish. The app reads the subject, the deadline, the hours available, the language and the weak topics. It asks one short follow-up if something is missing.



New session with one-tap ideas.

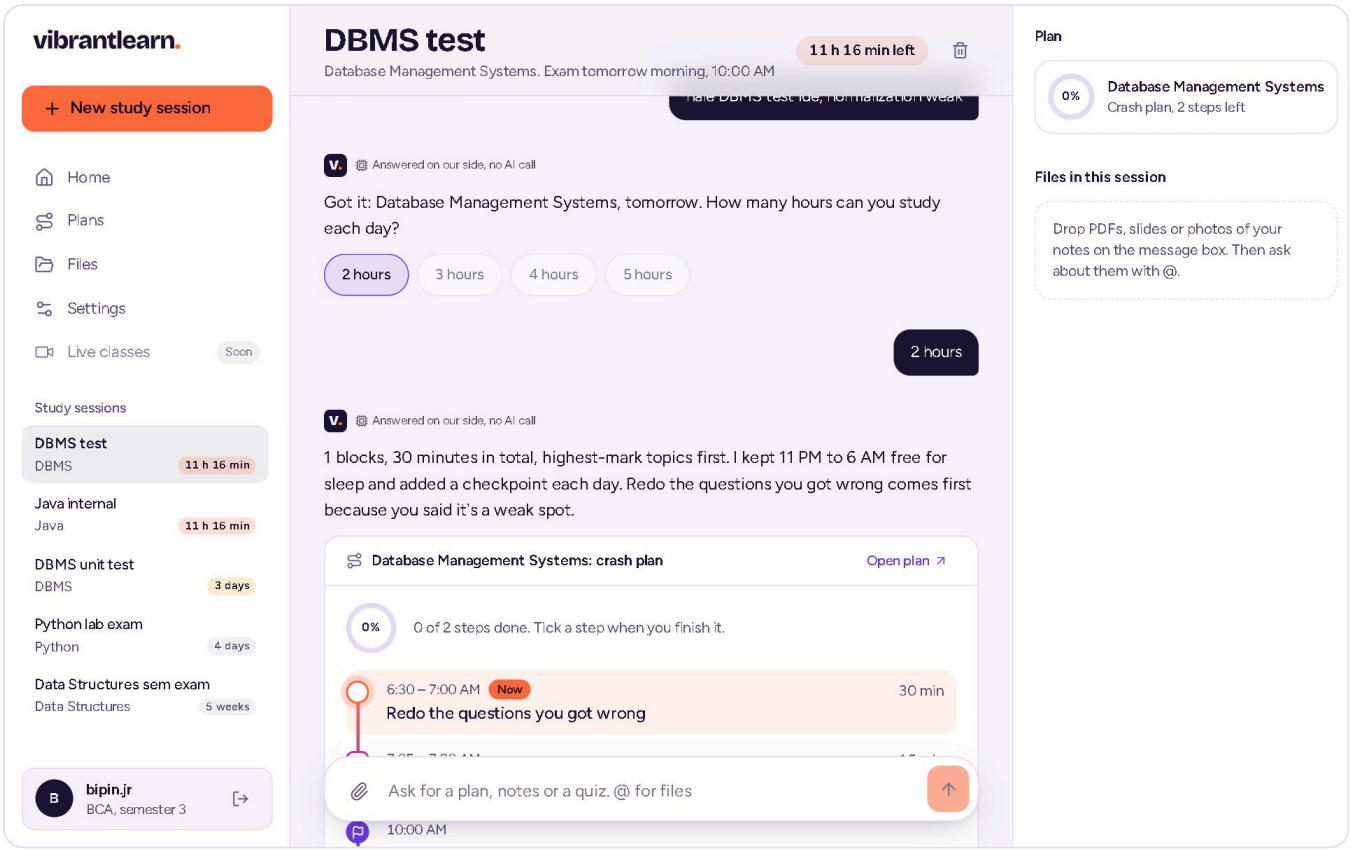


"nale DBMS test ide, normalization weak" is understood as DBMS, tomorrow. It asks for hours.

The student writes	VibrantLearn understands
Java internal tomorrow 10am	Java Programming, tomorrow 10:00 AM, crash plan
nale DBMS test ide, normalization weak	DBMS, tomorrow (Kannada "nale"), normalization first
kal OS ka exam hai	Operating Systems, tomorrow (Hindi "kal")
python lab exam on friday	Python Programming, the coming Friday
data structures sem exam in 5 weeks	Data Structures, five weeks, steady plan
explain deadlocks in kannada	Operating Systems notes, Kannada
overriding bagge swalpa heli	Java notes; answers in Kannada (Kannada typed in English letters)
DBMS 3 dinadalli exam ide	DBMS, in 3 days ("3 dinadalli")
ನಾಳೆ ಬೆಳಿಗ್ಗೆ ೯:೩೦ ಜಾವಾ ಪರೀಕ್ಷೆ	Java, tomorrow 9:30 AM, typed in Kannada script with Kannada digits
कल सुबह १० बजे जावा परीक्षा	Java, tomorrow 10:00 AM, typed in Hindi script

The app notices which language the student is writing in (Kannada or Hindi, in its own script or in English letters) and answers in that language. This step runs on our side without any AI call, so it is instant and free. The parser is covered by automated tests on more than 80 phrasings.

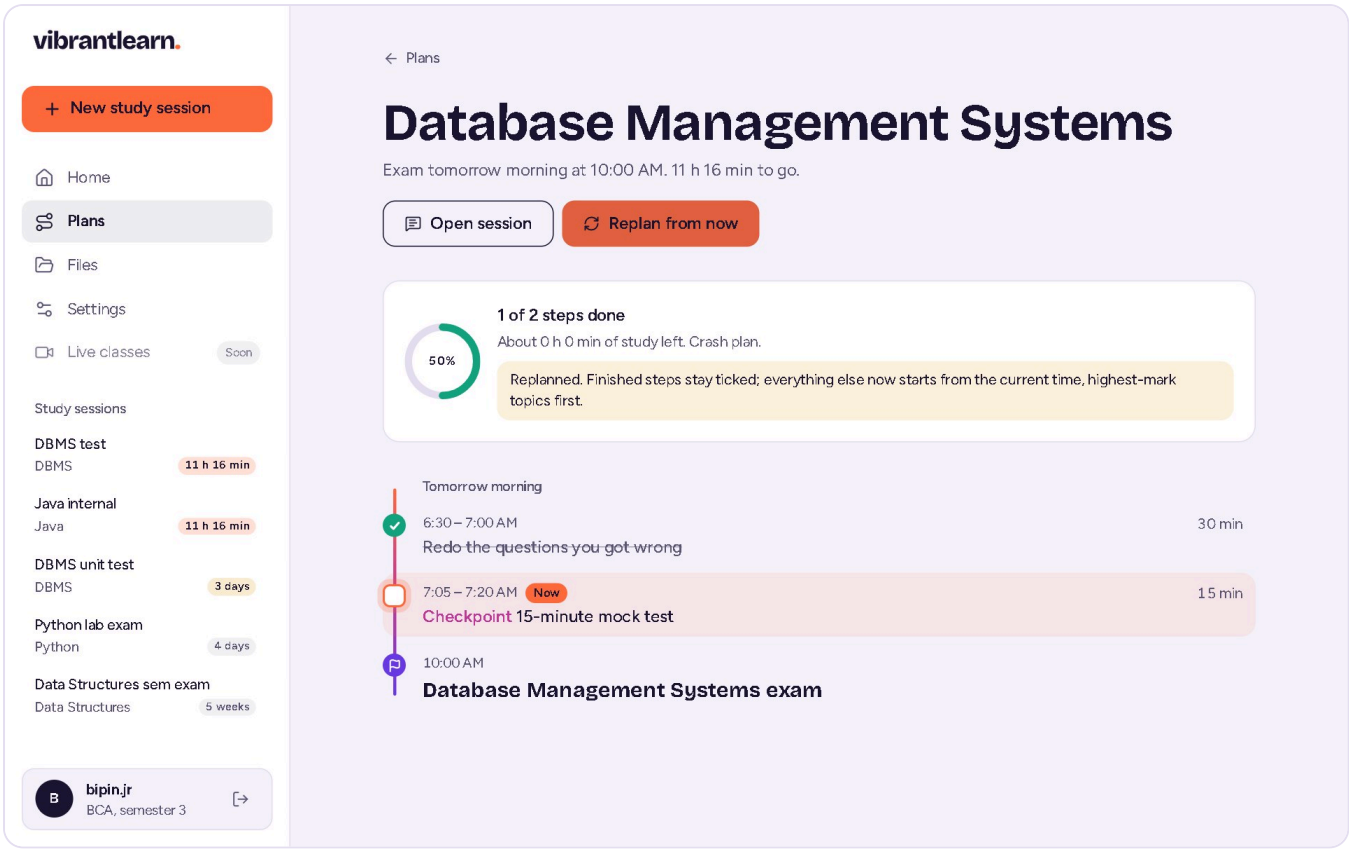
5. Personal study plans



A plan built in about a second: the weak topic first, real clock times, sleep kept free, checkpoints and the exam at the end.

Time left	Plan type	How it is built
Up to 6 hours	Sprint	Top-mark topics only, 25-minute focus blocks, a 15-minute mock quiz at the end
Up to 3 days	Crash	Evening blocks within your daily hours, a checkpoint every day, revision on the last evening, a short morning slot before the exam
Up to 4 weeks	Steady	A unit per week, weekend checkpoints, the last week for revision
Longer	Deep	All units, weekend checkpoints and three spaced revision loops

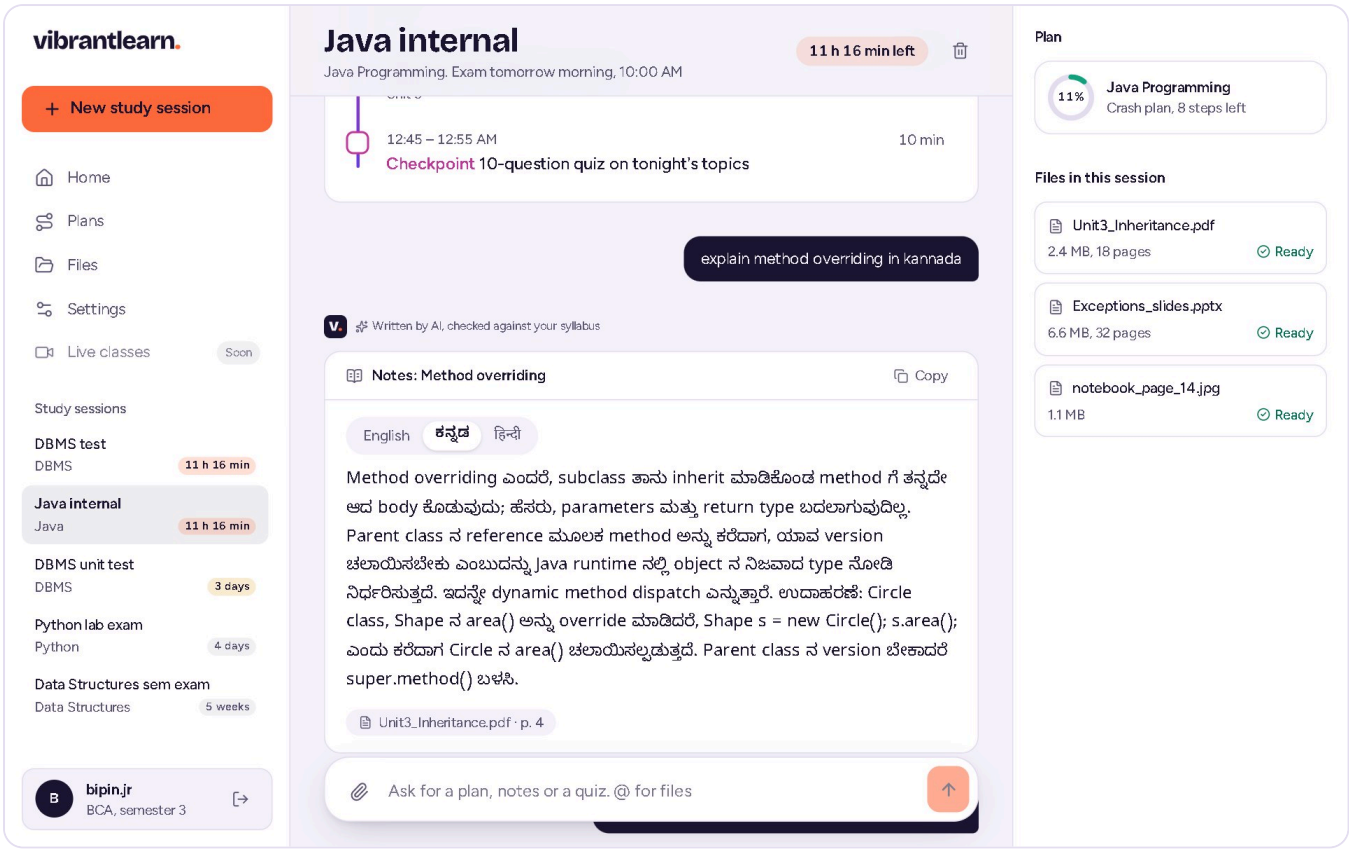
- Topics are ordered by **exam weight from the syllabus**, with anything the student called weak moved to the front.
- A plan **never runs past the exam** and stops at 11 PM.



Plan view: tick steps off, watch the progress ring fill, and replan from now if you fall behind.

- **Route-line timeline:** current step in Ember, finished steps in green with a tick, the exam in Violet.
- **Ticking a step** draws the tick, plays a short burst and updates the ring.
- **Replan from now** keeps finished steps and rebuilds the rest for the time actually left.
- The app warns when steps are **behind schedule**.

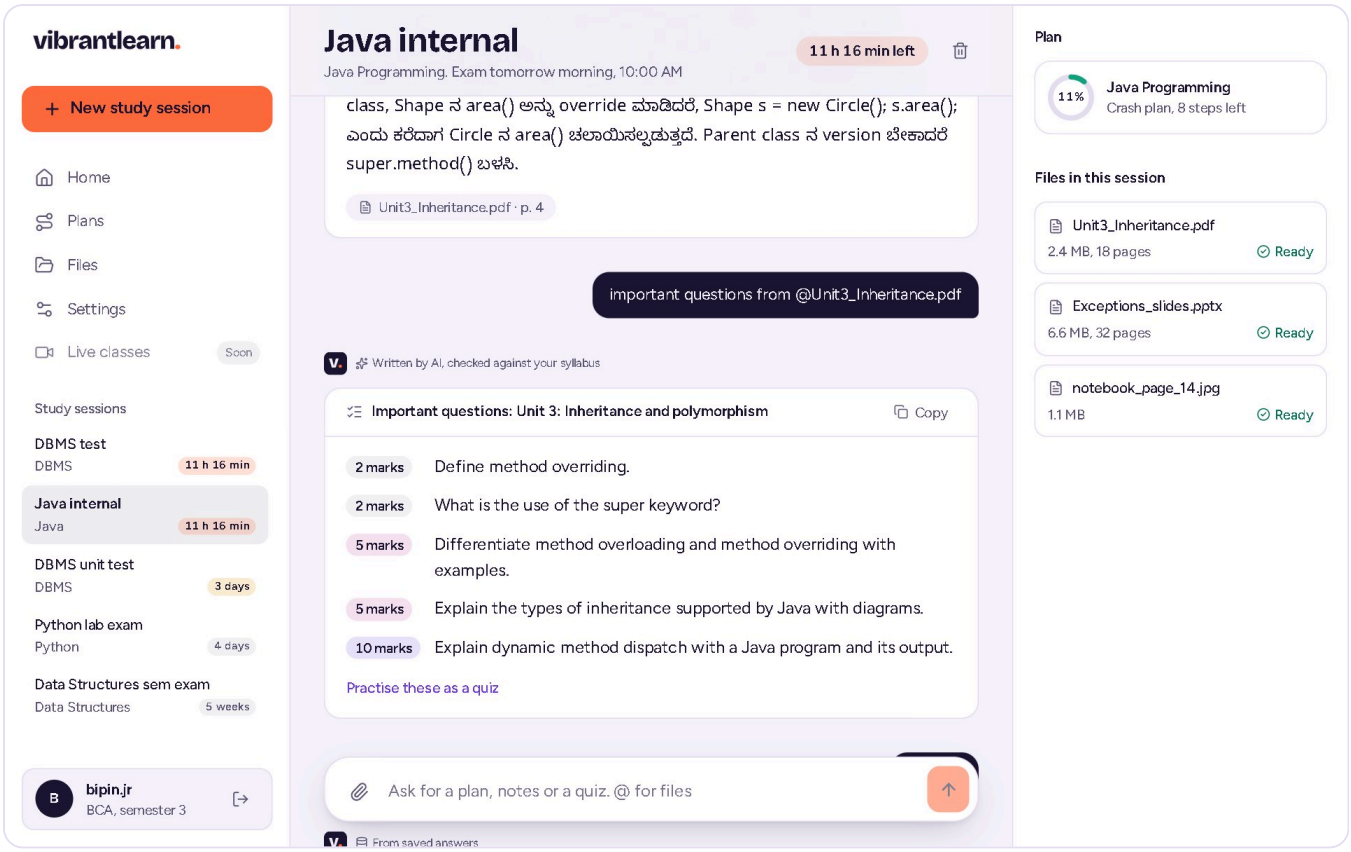
6. Notes in three languages



Short notes on method overriding in Kannada, technical words kept in English, with the source page from the student's own PDF.

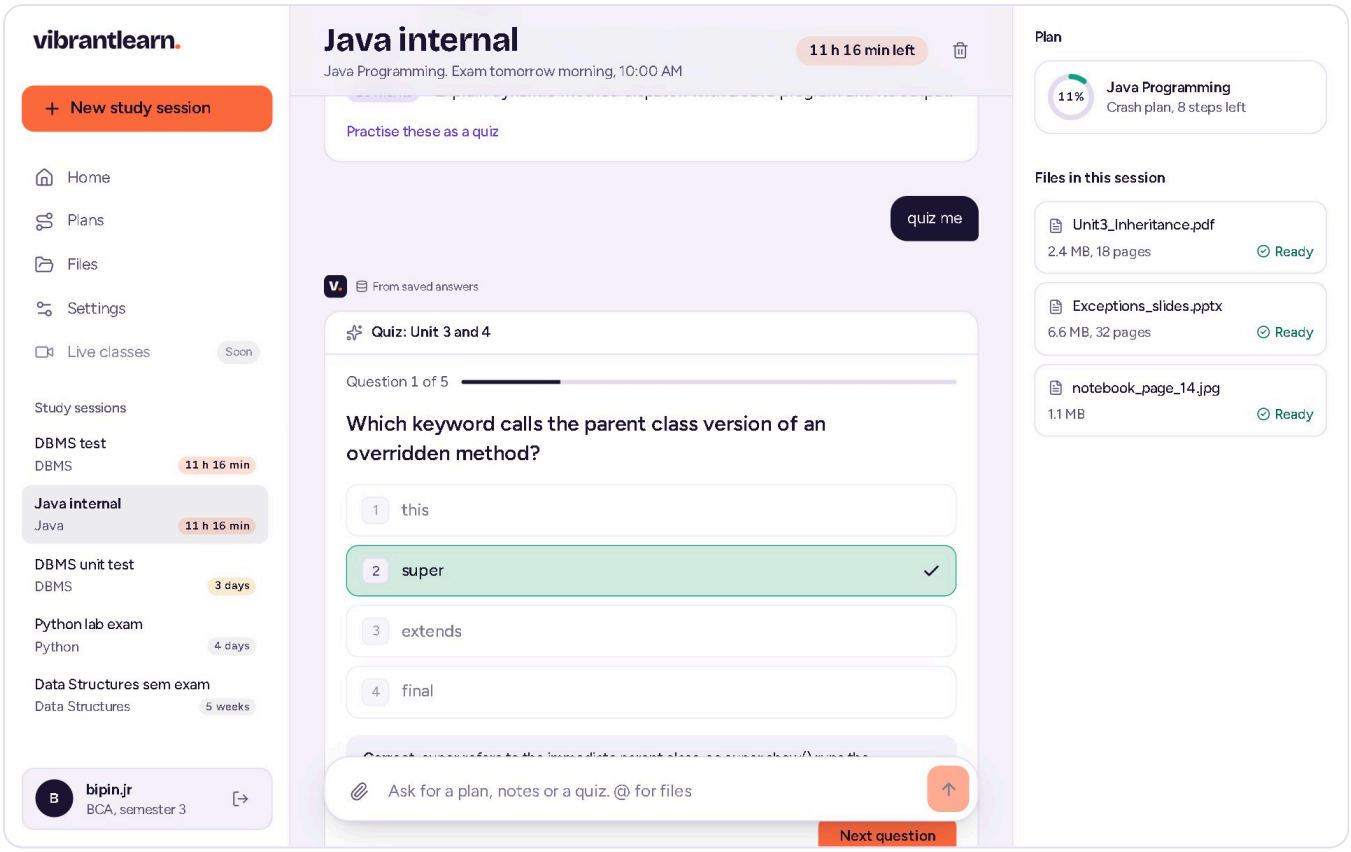
- Switch between **English, ಕನ್ನಡ and हिन्दी** on any note.
- Proper Kannada and Devanagari fonts.
- Copy button, and source chips when an answer came from an uploaded file.
- Notes for every syllabus topic are **written by AI in three steps**: write the English note, have a strict "examiner" check and correct it, then translate that exact text, keeping technical terms in English.
- Repeat questions are answered **from saved answers**, without calling the AI again.
- The whole app (menus, buttons, labels, dates) can be switched to **ಕನ್ನಡ or हिन्दी** in Settings.

7. Important questions



2, 5 and 10-mark questions in the university's style, with "Practise these as a quiz".

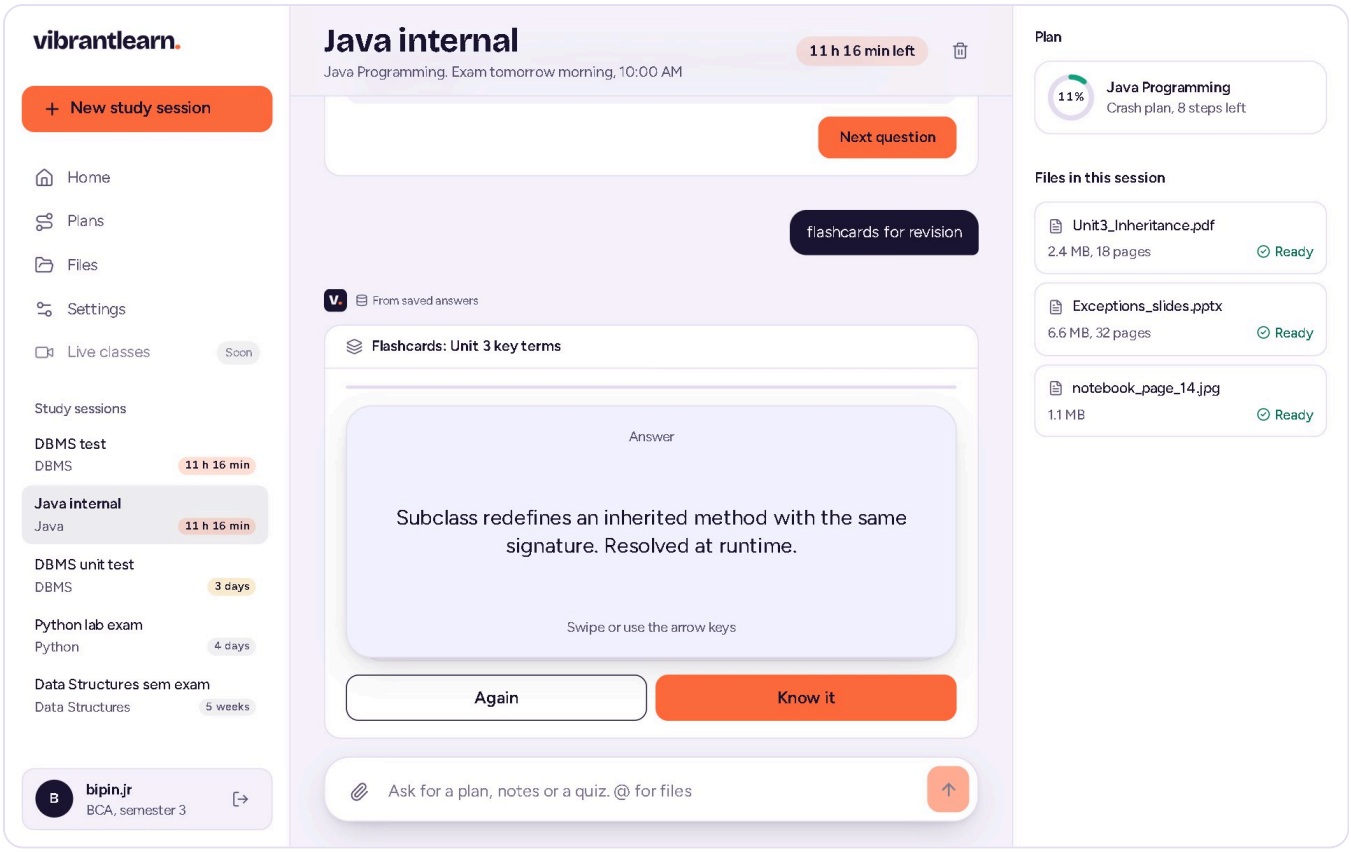
8. Quizzes



One question at a time. The correct answer fills green from where you clicked, and an explanation opens below.

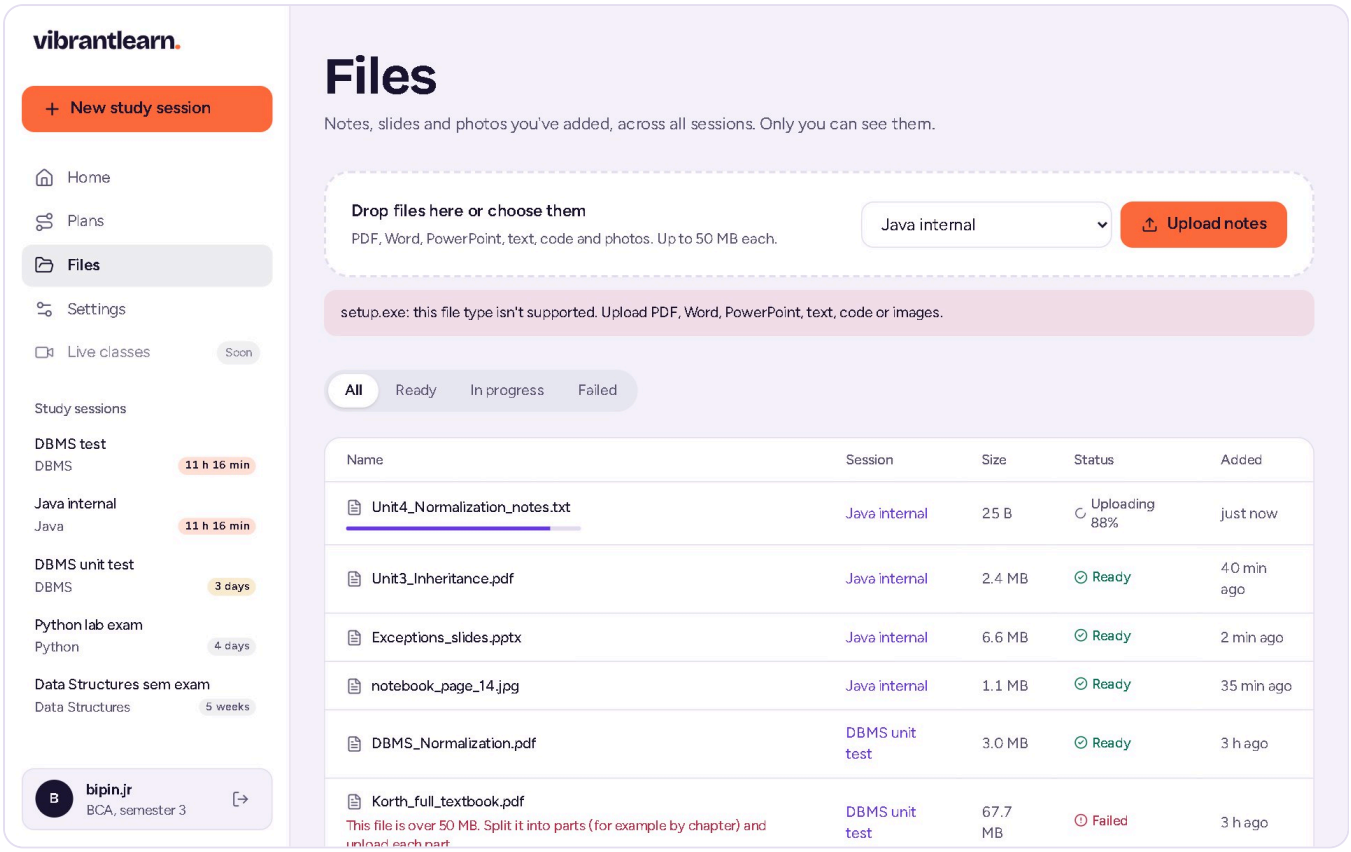
- Answer with the mouse or keys **1 to 4**.
- A wrong pick shakes and gets a red outline, and the right answer is shown.
- The score screen lists the **topics to revise**, with "Explain what I missed".

9. Flashcards



Cards flip in 3D. Swipe right or press → for "Know it", left or ← for "Again".

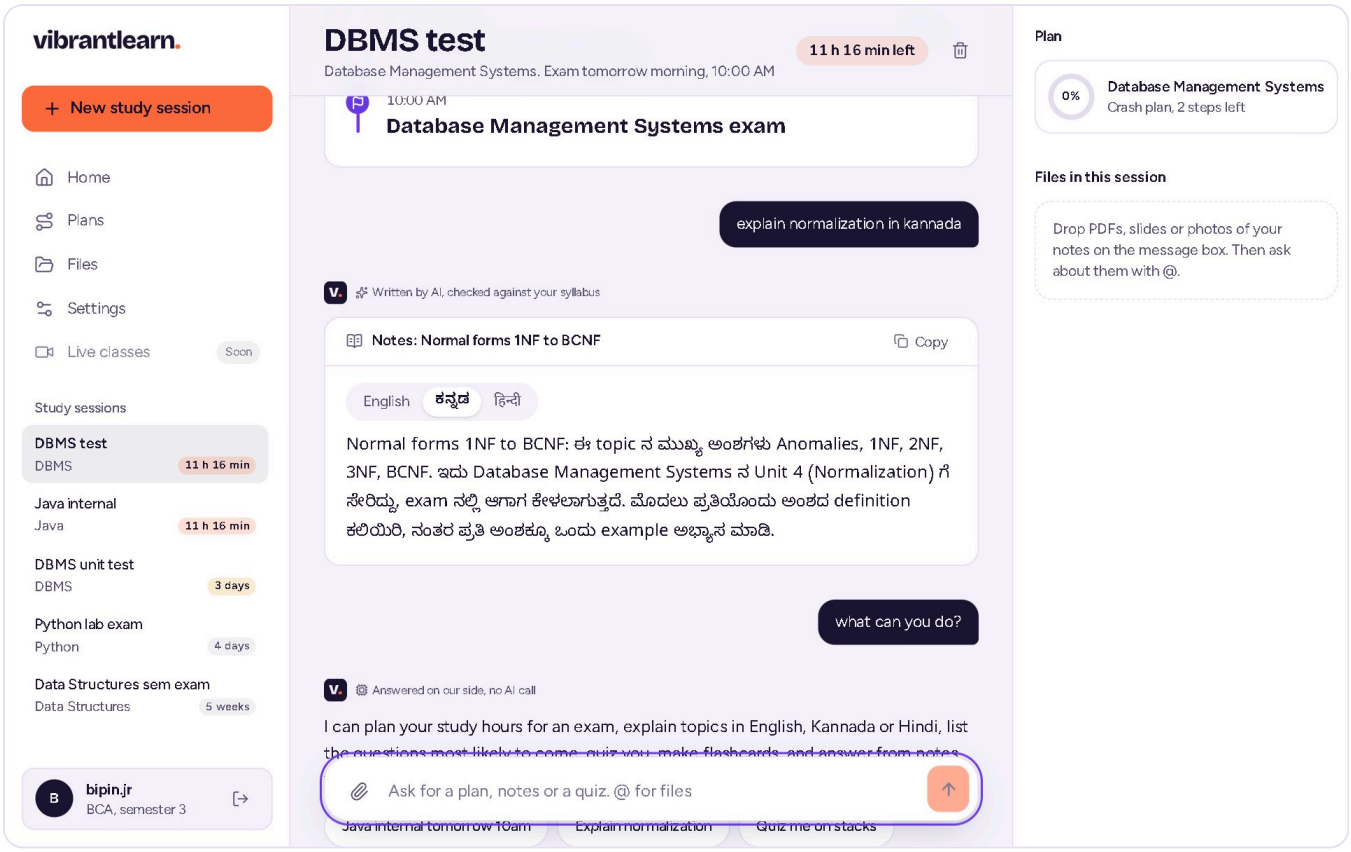
10. Your files



Uploads show real progress, then Checking, Processing and Ready. Unsupported files are refused with a clear reason.

- **Accepted:** PDF, Word, PowerPoint, text, code files and photos, up to 50 MB each.
- **Refused with a reason:** programs such as .exe, and files over 50 MB ("Split it into parts, for example by chapter").
- Ask about a file in chat with `@filename`. Answers show the file and page they came from.
- Drag files onto the message box or the Files page.

11. A chat that handles any message

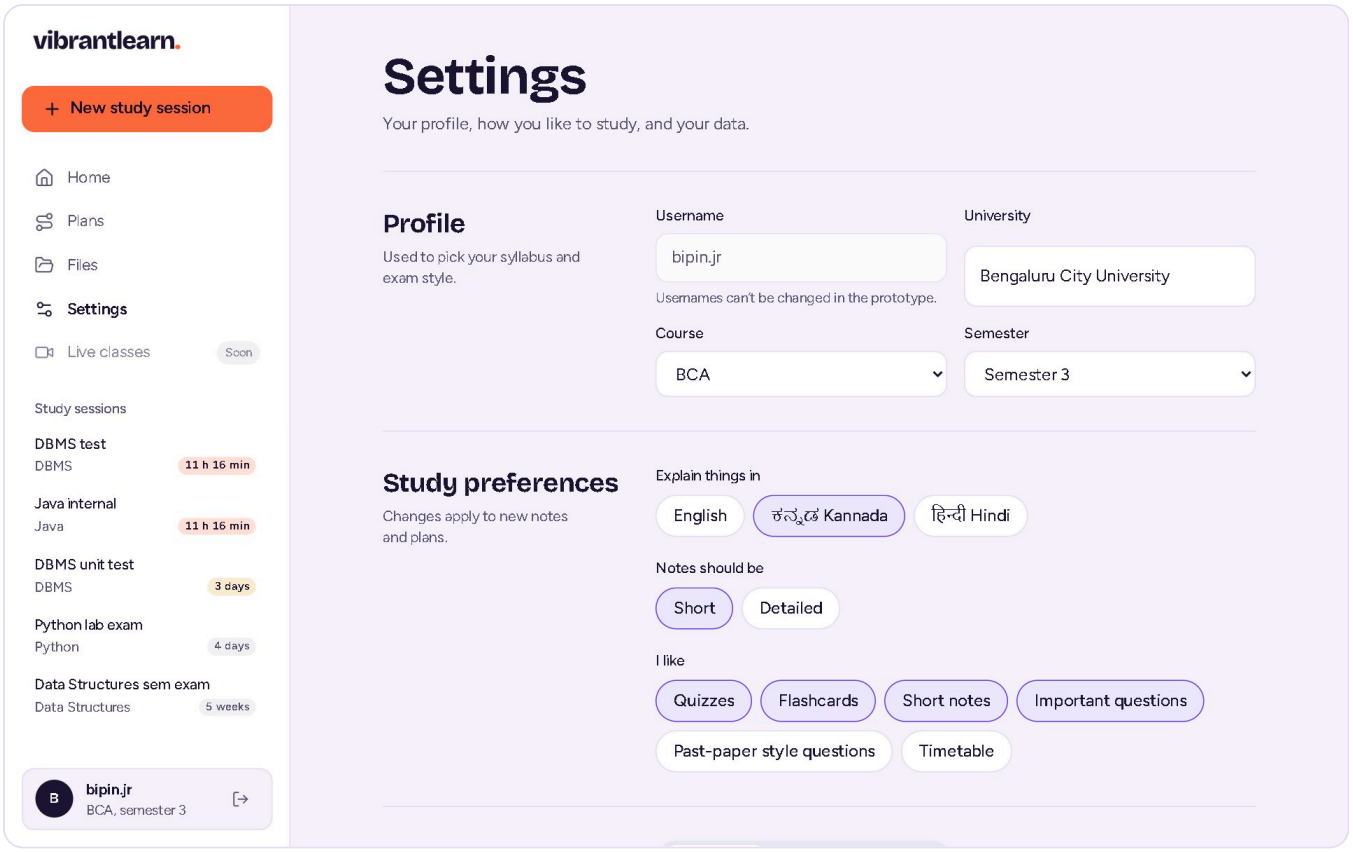


Study replies are documents (plans, notes, quizzes), not chat bubbles. Every reply says where it came from.

Message	What happens
Study requests (plans, notes, questions, quiz, flashcards)	Answered instantly on our side from the syllabus
"hi", "thanks", "what can you do?"	A friendly reply with suggestions to tap
Anything else ("what is photosynthesis", "how to stay focused")	Sent to the AI model (Groq) on the server, in the student's language. Without an AI key, a polite note and suggestions
Something unclear or gibberish	"I didn't catch that" with suggestions
Changing topic mid-question	The app follows the new request instead of getting stuck
Offline, timeout or AI busy	A clear message and a "Try again" chip. Nothing breaks

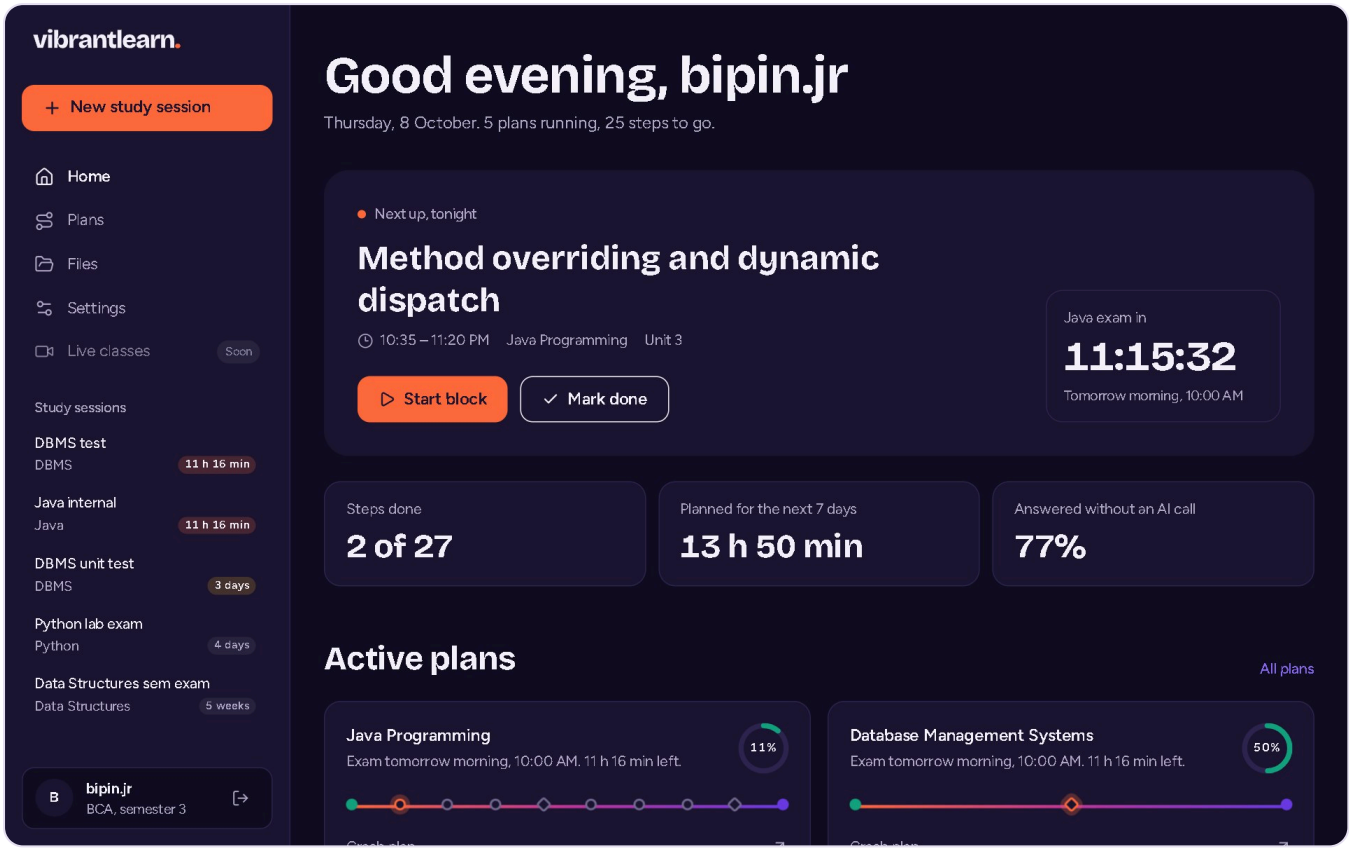
> To let the chat answer any question, put a free key from console.groq.com in `apps/web/.env.local` as `GROQ_API_KEY=...` and restart. The key stays on the server and never reaches the browser.

12. Settings and your data



Profile, study preferences, theme, usage limits, data export and account deletion with typed confirmation.

- **Study preferences** (language, short or detailed notes, favourite formats) apply to new answers.
- **Theme:** System, Light or Dark. Dark mode is designed, not inverted:

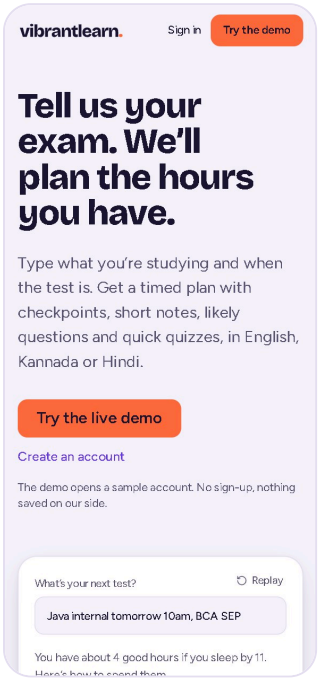


Dashboard in dark mode.

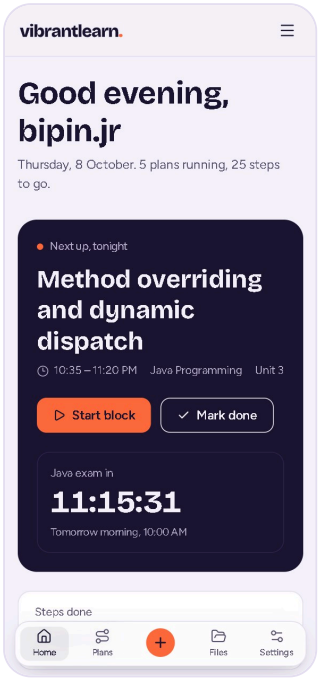
- **Usage meter** shows messages, plans and uploads against fair-use limits.
- **Export my data** downloads everything as a JSON file.
- **Delete account** needs the username typed (switched off for the demo account).

13. Phone, tablet and laptop

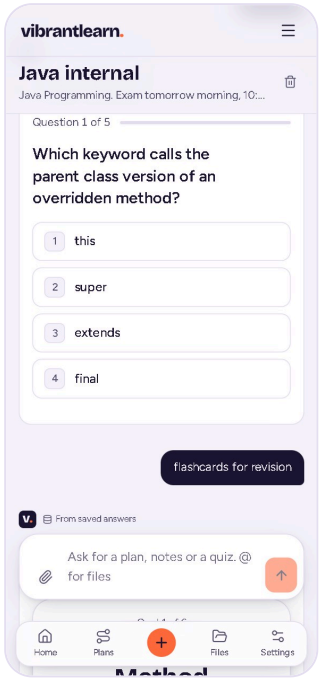
Every screen was built for phones first and checked automatically at 360 px (phone), 768 px (tablet) and 1280 px (laptop), with no sideways scrolling on any page.



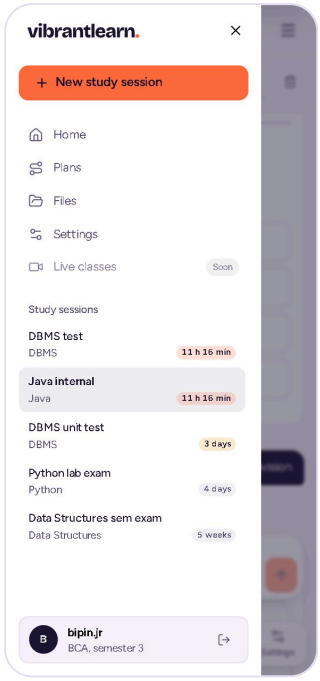
Landing



Dashboard and bottom bar



Study session



Slide-in menu

- Bottom tab bar with a central "New" button that's easy to reach with a thumb.
- The menu slides in from the side; Esc or a tap outside closes it.
- Touch swipes on flashcards; keyboard shortcuts on laptops.

14. Security and privacy

- **Data rules in the database (row level security):** each student can read only their own sessions, plans, files and quizzes. 34 automated database tests are written to check that one student can never see another's data; 36 of 36 pass on the live cloud database.
- **Private file storage:** each student's files sit in their own folder. Unsupported types and oversized files are refused, and files are never opened in the browser as web pages.
- **Browser protection headers:** a strict Content Security Policy with a fresh code per request, no framing by other sites, and no server version shown.
- **No secrets in the browser.** AI and database keys live only on the server.
- **Secure sessions:** the demo session cookie can't be read by page scripts.
- **The demo can't be abused:** demo mode is always off on the live website.
- **Fair-use limits** on messages, plans, uploads, sign-ups and the waitlist.
- **Usernames** are checked three times: in the browser, on the server and in the database.
- **Privacy** (India's DPDP Act 2023): a privacy notice, consent, export and delete. Files and chats never train any AI model.

15. How it was tested

Check	Result
Automatic end-to-end run in Chrome (17 steps: sign-in guard, landing, demo, plan, notes, quiz, flashcards, replan, uploads, settings, 3 screen sizes, sign-out)	All passed, no browser errors
Web unit and component tests (parser, scheduler, chat replies, quiz, formats)	84 passed
API tests (sign-in tokens, usernames, profile, waitlist, rate limits)	33 passed
Accessibility scan and colour contrast (WCAG 2.2 AA)	No issues; text colours at 4.5:1 or better
Type checking, lint and production build	Clean

16. What comes next

Next step	What it adds
Host the Python API and deploy	The live website, with AI and uploads served from the API
Native-speaker review	Kannada and Hindi text checked by native speakers
Read uploaded files on the server	Text from PDFs, slides and photos, searchable and quoted in answers
Verify the syllabus	Check the seed topics against the official BCU SEP syllabus
Later: teacher side	Small live classes, payments, moderation and notifications

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